

TLHEP¹ Portfolio

Raphael Hirschi

July 2011

¹Teaching and Learning in Higher Education Programme (TLHEP)

A teaching portfolio submitted for the full postgraduate certificate level
(2010-2011 handbook).

Dr Raphael Hirschi
School of Physical and Geographical Sciences
Keele University
Tel 01782 733324
Email r.hirschi@epsam.keele.ac.uk

TLHEP mentor:
Prof R. D. Jeffries,
School of Physical and Geographical Sciences

TLHEP tutor:
Dr E. Dawson,
Learning and Professional Development Centre

Contents

1	Introduction	1
1.1	Personal experience and pre-conceptions of learning and teaching	1
1.1.1	Personal experience of learning	1
1.1.2	Pre-conceptions of teaching	2
1.2	Role at Keele and modules taught	4
2	Learning	7
2.1	Learning levels	7
2.2	Constructivism	8
2.3	Approaches to learning	10
2.4	Experiential learning, reflective practice	11
3	Effective Teaching	15
3.1	Motivation	15
3.2	Constructive Alignment (CA)	20
4	Design	23
4.1	General context of the physics degree at Keele	23
4.2	ILOs at the module level	29
5	Delivery	33
5.1	Teaching and learning activities (TLAs) in physics/astrophysics at Keele	33
5.2	Good practice in delivering lectures	36
5.3	Interactive teaching and active learning	40
5.4	Lectorials	43

CONTENTS

5.5	PowerPoint or white board or both? What else?	47
5.6	Handouts and textbooks	48
6	Formative and summative assessments	53
6.1	Assessment in physics/astrophysics at Keele	53
6.2	Formative assessment	55
6.3	Summative assessment	58
6.3.1	Feedback on exam papers	60
7	Conclusions and Outlook	63

1 Introduction

1.1 Personal experience and pre-conceptions of learning and teaching

1.1.1 Personal experience of learning

Throughout school and high-school, I had to study many diverse topics. I found scientific topics easy to study, liked them and did well at them and found topics like history or literature hard and not so interesting. Concerning history, I did not see any connection between what I was learning and my life. This is in part due to my environment since I grew up in the country side, where world affairs were rarely discussed. However, I do not recollect my teacher trying to link what we learnt in history to our own life or how learning about history can help us create a better future. It is only at university that I had this type of discussion with a friend and since then my interest in history has increased because I understand, rather accept, its relevance. In literature, apart from getting bad marks, I was never given feedback that allowed me to improve my composition skills. Furthermore, as a student, I was never clearly taught about how one learns and the different learning levels (Bloom's taxonomy) that a student is expected to achieve in the context of the topic learnt. The only advice on how to learn that I remember was from my maths teacher at high school, who told us: "we listen and forget, we write and remember, we do and understand". I always kept this advice in mind by taking notes during lectures and revision and I also know that using PowerPoint in lectures has the disadvantage that students do not take notes and therefore it is harder for them to remember what was taught during the lecture. I think that these 3 aspects: relevance (linked to motivation), effective feedback and help on how to learn, could

have helped a lot to improve my learning in topics I struggled with. Now that I have realised this, it certainly helps me to view struggling students in my class from a different point of view and my challenge is to help them in the same way I had hoped my teachers would have helped me. This is one of the reasons that motivated me in the next chapter to provide examples for Bloom's taxonomy in the context of the module that I teach and I plan to hand out these examples along with a copy of Bloom's taxonomy to my students in the future.

As I learnt during the TLHEP, it is important to know what students are doing during lectures. So I looked back at what I was doing during my lectures when I was a student. For scientific topics, I was continuously following derivations of theorems and trying to figure out the meaning of what was being taught. I was following the lectures so closely that I could spot most mistakes the lecturer was doing on the board. In short, I actively participated in my lecture. I think this helped me a lot to achieve higher learning levels and as I will explain in this portfolio, I design my lectures to promote as much as possible the active participation of the students (see Sect. 5.3 and Sect. ??).

1.1.2 Pre-conceptions of teaching

Before starting on the TLHEP, the only way to design and prepare for my teaching was to look at what my teachers or colleagues have done or to look at textbooks. I did not have the theoretical framework in order to understand student learning, explain how what I teach helps the student learn better or judge how effective my teaching is. The course thus represented a steep learning curve for me but before I present what I learnt during the course, it is interesting to discuss what I knew about teaching before the course and what I thought was good practice.

I thought the best method to teach scientific topics is first to explain as clearly as possible the theory, then provide examples and finally ask the students to solve problems. This is the method used predominantly around the world and thus it must be a good method but how can I be sure it is working? Since I did not know about levels of understanding, intended learning outcomes or constructive alignment, the only answer I could pro-

vide is that it works because we check the students' understanding by asking them to solve problems and they cannot solve the problems without a good understanding of the theory. Although this answer is broadly correct, I cannot use it to evaluate and improve my teaching. In this portfolio, I will for example discuss how I used the theory that I have learnt during the course to check that my exam papers are aligned with my goals for the modules (by linking sections of the exam papers with levels of understanding in Sect. 6.3).

I thought that good practice for scientific topics was to teach using a deductive method (teacher explaining to the student in the best possible way the theory) because an inductive method (asking the students what they know about a topic and developing the teaching around what knowledge the students contributed) is confusing. I thought it to be confusing because students would provide incomplete explanations and even totally wrong ones. I will explain later how my view has changed on this point in Sect. 5.3.

I thought that since it is the students' choice to be here, they are motivated so I do not have to care or worry about motivation. As long as I explain the material as clearly and nicely as possible to help the student accommodate it, I am doing my job. Having reflected on my own experience of learning about history, I think that it helps the students if examples and images used are more interesting and varied because it will help many students to want to know more about a topic rather than reluctantly learn some topic. By making the topic relevant to their life, it will help motivate them and also improve their learning as I will explain later. I discuss this point in Sect. 3.1.

Finally, I thought that the students choose their approach to learning. Obviously, students come with varied experiences and environment, which would influence the way they approach learning. I did not realise that we, as teachers, can affect how they learn to a great extent. A particular example that I will mention later concerns surface approach to problem solving in my mechanics module (in Sect. 5.5). The first year I taught that module, I provided the answers to the questions to the students. Some students therefore naturally read the answers (surface approach) instead of trying to

find out the answer themselves (deep approach). The next year, I prevented a surface approach by blanking out the answers on my lecture slides. This was really a crucial incident that totally changed my view on how much what I do as a teacher affects how students learn during the class. Now I know that we can encourage or even force deep learning almost independently of the student's motivation. This is further discussed in Sect. 2.3.

1.2 Role at Keele and modules taught

I arrived at Keele in 2007 as an RCUK academic fellow. This was a research fellowship but also intended at training the fellow as an academic. I therefore started teaching two modules during the fellowship.

The first module is a “mechanics” module part of the physics Science Addition Specialism Programme (SASP). In this portfolio, I will refer to this module as the **SASP** module. The participants on this programme are practising school teachers who want to improve their physics knowledge or who want to start teaching physics up to A-level. The level of material taught is thus equivalent to a first year undergraduate programme. However the context is very different since the students have a full-time teacher's position in a school and only come to Keele every Wednesday. This means that the teaching is more spread out, usually only one lesson per month, which means they usually forget what we did the previous time and we have less contact time together. Furthermore, most students have not studied for (many) years when starting the programme. Since I teach one of the first modules of the programme, the students are not only new to my module but to the entire programme and getting used to study again. These students can get demotivated quite easily and I discuss this point in Sect. 3.1.

The second module is also “mechanics” but is part of a different programme, the subject knowledge enhancement (SKE) course, previously known as the physics enhancement course (PEC). The students on this programme are people who want to become school teachers. Thus many students have just finished a degree (not physics) or have decided to change career. The intake of students is thus very varied with some students knowing most of what I teach and others who do not master the necessary pre-requisite

skills, in particular mathematics. This is a additional difficulty for the implementation of lectorials as discussed in Sect. 5.4. This module is also at the start of the programme so students have to adjust to their environment but since they are full-time students, lectures are frequent and they settle in much faster than the SASP students.

In November 2010, my fellowship ended and I was appointed as a lecturer. This academic year I have still been teaching the two mechanics modules and also been teaching a first year, second semester module on “Electricity and Magnetism”. I will refer to this module as the **EM** module. Although this is a first year module, students have completed the first semester and have thus settled in their environment (unlike the SASP students). They have also completed a “mathematics” and a “mechanics” module in the first semester (that someone else is teaching) so they have ALL already learnt concepts that I can refer, compare, contrast to. As discussed in this portfolio, relating new concepts (schemata) to previously known ones is an extremely common practice in teaching in science and is supported by constructivist theory. The EM module runs in parallel with the “oscillations and waves” module, where students learnt some of the concepts that I teach in my module so I can also build bridges between the two modules in order to improve their learning.

2 Learning

Since the goal of our teaching is for students to learn, it is natural to first look into how students learn. This chapter is intended as a brief overview to the topic of learning and I have tried to use as many examples as possible out of my own teaching to relate it to my personal experience but the link between learning theories and my own practice will continue throughout the rest of the portfolio.

2.1 Learning levels

Learning is not that easy to define. It is about memorising new facts, understanding new concepts, being able to apply a theory, creating new theories, The different aspects of learning require different levels of understanding and cognitive abilities. Bloom's taxonomy classifies different aspects of learning according to levels of understanding and is often used to establish intended learning outcomes (ILOs) for teaching (see Sect. 3.2).

Bloom's taxonomy (1979) covers six levels of cognitive ability, increasing from knowledge to evaluation. I explain here the taxonomy in the context of the equation of the electric force between two point-like charges, q and Q ,

$$F_e = \frac{kqQ}{r^2}$$

, where k is a constant and r is the distance between the charges. I use this equation because it is the first equation that I teach in the EM module. The six levels are:

Knowledge Students memorise the equation and could write it down in an exam but might not even try to understand whether it makes sense.

The student might not have thought it was strange if the force had not been proportional to one of the charges.

Comprehension Students understand what each symbol means and relates for example the fact that the electric force is a force between two charges and it is thus logical that the force is proportional to qQ .

Application Students are able to use the equation to calculate the answer of a specific problem.

Analysis The students are for example able to determine the impact on the magnitude of the force if the charges are moved further apart or closer to each other and understand what happens at infinitely small or large distances.

Synthesis The students are able to understand the equation in the general context of forces, able to compare the electric force to the gravitational or magnetic force, understanding similarities and differences and able to solve problems where several types of forces are at play.

Evaluation The students understand the hypothesis, validity and limitation of the equation. In the EM module, another example is to compare two laws (Biot-Savart and Ampère's laws) in order to determine the magnetic field and explain which law is easier to use depending on the situation.

This small exercise also shows that in order to achieve a certain level, previous levels must also have been achieved. Students cannot apply the equation unless they have learnt it and understood it. As I mentioned in the previous chapter, I intend to distribute Bloom's taxonomy along with the above examples in order to help my student recognise the different levels of understanding and what in practice they should do to achieve higher levels.

2.2 Constructivism

After briefly exploring what learning is, it is important to know how learning takes place. The most widely used theory of learning is constructivism.

In this theory, knowledge is built on previous knowledge, in the sense that new knowledge is assimilated by modifying and expanding existing concepts/understanding, called schemata. In the modules that I teach, mechanics and electricity and magnetism (EM), most students have already learnt about both topics at A-levels but they have to expand and also modify their understanding of these topics. One particular example is direct current (DC) circuits, which the students have already studied and alternating current (AC) circuits that they are learning for the first time. They are similarities between the two but also importance differences that imply that the schemata that the students use to understand DC circuits have to be modified in order to understand AC circuits. It is very helpful that the students have already learnt about a topic to a certain level and teaching in physics continuously uses this fact by drawing parallels to previous knowledge from earlier modules. When presenting a new topic, most of my lecturers spent time to compare and contrast with known concepts. I also do it in my electricity module by relating the electric force and potential to the already studied concepts of gravitational force and potential. In physics, the curriculum is thus designed in a spiral manner ([Bruner, 1960](#)), revisiting each topic to a higher level at each iteration. However, I had up to now not appreciated that the reason for these constant comparisons to previous concepts is directly linked to how students learn. Another aspect as important as comparing to previous knowledge is clearly highlighting the differences between new knowledge with previous one in order to enable the existing schemata to be modified (see [Fry et al., 2009](#), page 10, who cite [Mezirow \(1991\)](#) as well as Sect. [2.4](#) for the related concepts of assimilation and accommodation). This is not as easy to do as highlighting similarities. In my EM module, it seems easy to list the similarities between the electric force and the gravitational force but I had to make a conscious effort to clearly explain the differences between magnetic fields and electric fields (different field lines, dependence on distance, differences between Gauss's law and Ampère's law).

2.3 Approaches to learning

Before starting the TLHEP, I thought that the way the students learn is up to their ability and motivation. This is a level 1 approach to teaching based on “who the student is” as explained in (Biggs & Tang, 2007). This was probably influenced by the Swiss school system that I followed, which is very selective and in which students are streamed into different levels from secondary school. In this way of thinking, different students have a better or worse ability to learn, which is hard to change. The ability of the students to learn in a certain way is often referred to as learning styles. I would personally prefer to call it a pre-disposition to a certain learning approach and use learning styles to differentiate between students who prefer to learn by doing (solving problem) from those who prefer to learn by reflecting on a topic. Fry et al. (2009, on page 18) list four main learning styles: activists, reflectors, theorists, pragmatists, (taken from Honey and Mumford 1982) in this line of thinking. The main disadvantage when focusing on learning styles or pre-dispositions is that we think that we cannot change them so it is entirely up to the students and their genes how well they can learn. Instead, as explained in Biggs & Tang (2007), by using the term approach to learning, we break the idea that the teacher cannot influence the way the students learn. Of course, the students still have some choice and responsibility in how they learn but we can design learning activities that can prevent or encourage different learning approaches. Another important aspect is that students’ pre-disposition to using a certain learning approach depends on many factors (ability, external constraints like family, background knowledge of a particular topic, ...), which are often time dependent. By focusing on learning approaches and learning activities that encourage good learning approaches, we can very often prevent a student from passing a module using bad learning approaches. I already mentioned the example of having answers of problems in my mechanics lecture notes in Sect. 1.1.2 that show how sometimes it is easy to prevent a bad learning approach and come back to this point in Sect. 5.5.

What are bad and good learning approaches? Most textbooks on teaching call good learning approaches “deep” and bad ones “surface” . Students

using a surface approach will focus on the “signs” of learning as Marton says (see for example [Biggs & Tang, 2007](#), chapter 2 and references therein). The students’ focus on the words or formula instead of trying to understand their meaning. Taking the example of a problem class in physics, students using a surface approach are mostly concerned about their final answer and often asked “Is my answer correct?”. On the other hand, students using a deep approach focus on the meaning of the theory they are using and ask questions about the method they are using or the theory they are applying. [Ramsden \(2003\)](#) and others consider two aspects of learning approaches. The first one concerns “what” the student learns, is also linked to the meaning of what is learnt and the two possible approaches are also “deep” and “surface” approaches. The second aspect concerns “how” students learn. This aspect focuses on how students structure and organise what they learn. The negative approach for this aspect of learning is “atomistic”. Students using an atomistic approach segment the knowledge they are learning and in the context of a problem class, fragment the questions not seeing links between the different parts or which part of the question is important for the current module. The good approach is called “holistic” and in this case the students integrate different parts with each other and with the general aspects of the module. In a problem class, students using a holistic approach think about the different parts of the questions and how they fit together. They also relate what they are doing in the problem class to what they learnt in the lecture and often will ask questions that are not part of the problems sheet. Using Ramsden framework, the approach we want students to use is a deep-holistic approach and we want our learning activities to prevent students from using surface-atomistic approaches.

2.4 Experiential learning, reflective practice

Although experiential learning, learning from experience, relates more directly to “learning on the job” situations than to university lectures, it is a very useful framework to use for reflective practice for lecturers. David Kolb (1984) developed the now called “Kolb Learning Cycle”, which is a model that describes the four different actions (or abilities) of the learning

cycle, that is continuously repeated in experiential learning:

- “Concrete Experience (CE)”
- “Reflective Observation (RO)”
- “Abstract Conceptualisation (AC)”
- “Active Experimentation (AE)”
- “Concrete Experience (CE)”
- ...

In my teaching, I went through this cycle without knowing about it. As I said in the previous chapter (Sect. 1.1.2), I thought that inductive teaching methods were not good methods to use in science because I thought that it was confusing. After learning more about inductive methods (asking the students what they know about a topic so contribute knowledge before I start teaching) and interrogating practice during the course of the TLHEP (entering Kolb’s cycle at the “Abstract Conceptualisation” step), I decided to try to use inductive teaching in one of my lectures (“Active Experimentation”). It turned out that the inductive method worked much better than I had expected (“Concrete Experience”). I then tried to understand why it went better than I had expected (“Reflective Observation”) and I discuss below why I think an inductive method is much better than I had thought. Now that I know about constructivism, I understand that learning never takes place on a blank sheet but instead learning is built upon previous knowledge. Piaget (1896 – 1980) developed the concepts of assimilation and accommodation in order to explain the learning process. These concepts are two complementary and inseparable steps of the learning process. Atherton (2011) describes these two steps as follows:

- ”Assimilation: *fit practice to theory*. Complex but familiar external objects are simplified to fit pre-existent categories in your head“
- ” Accommodation: *fit theory to practice*. You have to change the ideas in your head to fit the realities of external objects”

In my teaching experience, using an inductive method helps the students activate their prior knowledge. This thus helps the assimilation process because the students are anyway trying to fit new knowledge (“practice”) into their existing knowledge (“theory”). I have also now realised that the best way to improve accommodation is to then compare and contrast the new knowledge (“practice”) that I teach with prior knowledge (“theory”) in order to help the students *“fit theory to practice”*.

After this “reflective observation”, I have now myself accommodated my knowledge of teaching and created a new “Abstract Conceptualisation” (AC) through my journey in the Kolb cycle. My next step is to go through a new cycle by again implementing what I think is good practice in teaching (“active experimentation”) and see how it works.

3 Effective Teaching

As Biggs & Tang (2007) summarise on page 31:

“Effective teaching requires that we eliminate those aspects of our teaching that encourage surface approaches to learning and that we set the stage properly so that students can more readily use deep approaches to learning. This involves getting the students to agree that appropriate task engagement is a good and impelling idea (otherwise known as “motivation”).”

Even though we can force students to do a certain task or sit in class, students enjoy the tasks much more and learn better if they are motivated to do so. Therefore one of the first goal of teaching is to motivate students to learn, which I discuss below.

3.1 Motivation

We wish all our students were intrinsically motivated (like Susan in Biggs & Tang, 2007, first introduced on page 8). After all, I, like most lecturers, was a very motivated students, with a great interest in the topic being taught. We could therefore say that it is the students’ fault if they are not motivated, which is a level 1 teaching approach (“What the student is”, see Biggs & Tang, 2007, page 16). This approach implies that students’ motivation is something that cannot be changed. I must admit that this is what I thought before starting this course (as mentioned in Sect. 1.1.2). The problem with this mindset is that many students are not intrinsically motivated (the Roberts in Biggs & Tang, 2007) and that we think we can do very little to change this. Over the course of the TLHEP, I have fortunately realised that motivation, like learning approaches, is much more variable than I initially thought and that it can be influenced by teaching methods. If I

look back at my own motivation, even though I was generally motivated, my motivation varied with the topic and also with time. In fact, more often than not, motivation follows understanding rather than the opposite. A good example concerning my hobbies is cricket. I did not know anything about cricket before moving to England and found it boring and uninteresting. However, after playing one or two seasons with friends, I have learnt most of the basics of cricket and now find it much more interesting.

Biggs & Tang (2007) give two main factors that make students want to learn something:

- “It has to be important; it must have some *value* to the learner”
- “The learner needs to *expect success* when engaging the learning task”

“Good” students, like Susan, both find the topic intrinsically important and interesting and think they will succeed in the task. They are therefore very motivated. On the other hand, “weaker” or less intrinsically motivated students will have a fragile motivation that can easily be affected by the teaching environment. In order to increase the “value” or importance of the topic, I now try to highlight the importance, relevance of the topics that I teach as often as possible during my lectures. Here are some examples:

- Instead of just having a plan on my title slide, I choose an image that will raise the interest of the students in the topic. As often as possible, I try to relate the topic of the day to every day life. For example, I chose a picture of a wind farm for the topic of electric generators. These connections to everyday life not only help motivate the students but also help the assimilation of new topics by linking them to what they already know. Other times, it is just a pretty image explaining the theory that I will present.
- During the lectures, I try to give as many every-day-life applications of the topic learnt. For the EM module, there are many such examples: light bulbs, radios, televisions, micro-waves and magnetically-levitated trains.
- I use topics from my own research in order to write more interesting problems sheet questions. One such example is to write questions on

the solar system, our galaxy and neutron stars, in my mechanics (SKE and SASP) modules.

I also try to increase students' motivation by using online java applets during my lectures, mostly taken from this website: <http://higheredbcs.wiley.com/legacy/college/cutnell/0471151831/concepts/index.htm>. These java applets are small animations, which bring to life the concepts covered and also provide a nice break from the usual talking and writing on the board. They are similar to experiments that professors brought into my classroom when I was a student but the main advantage is that the students can use the applet at their own convenience since they are available online. In these applets, the students can vary key quantities and see the impact it has on the system (for example change the resistance of a resistor in a circuit and see how the current is affected). It therefore helps them comprehend and apply the knowledge taught in class with instant feedback. Up to now, I have used free online resources because the general textbook that we use for first year students does not have accompanying online resources but there are other textbooks with integrated online resources. I would prefer to use a textbook with online resources for the reasons given above but it is not easy to change the recommended textbook for first year physics students. First, we must find one book that is calculus-based since this is part of the skills that we want our students to develop and many textbooks are algebra-based. Second, we must find a textbook that covers as many first year topics as possible to make it worth for the student to purchase the book. Third, we want to make sure that the students care about online resources before committing to a new textbook. I therefore asked the students during my module to try out the online resources that I had been using and was hoping to get feedback out of the evaluation forms that the students.

Unfortunately, I found out that getting useful feedback out of students is not as easy as I had thought. For the EM module, I had to hand out a standard form supplied by my department so did not prepare a separate feedback form. Since the standard feedback form did not have a specific question about online resources, only one student mentioned that one of the best points was "The quantity of online resources available" (response 18).

Since I did not ask the question directly, it is a good sign that one student mentioned it but I would have liked to have the opinion of more students. At the other extreme, for the SASP and SKE “mechanics” modules, I prepared a very detailed feedback form (see appendix ??). Since the feedback form was very long, I asked the student to fill it in at home because it would have taken too much time during the lecture. As a results very few students returned my feedback forms and I do not have much more information than with the standard feedback forms of the EM module. In the future, I have to ask only a few specific questions so that I can have specific feedback from enough students or just asked the students in class rather than using feedback forms in order to get feedback on specific aspects of the module. Since writing this paragraph, I have actually already used my improved skills at obtaining feedback. I was teaching at a two-day training event for school teachers and was able to obtain valuable feedback that we will use for our Research Excellence Framework evaluation. I did this by adding only two very specific questions to a standard questionnaire and thus all the students completed the feedback form. Concerning the use of online resources, I had a discussion with Rob Jeffries and Dean McLaughlin who have also tried to use online resources in their modules. Dean McLaughlin checked the traffic on WebCT and found that for his as well as for and my module there were less than 50 clicks on the online resources link, whereas there were about 5000 clicks on the module pages. This means that students are not as keen as I thought to use online resources, which is surprising for me but I have to accept the reality. Rob Jeffries and Dean McLaughlin told me that screencasts are popular so I am considering using screencasts or creating videos of my lectures to help students learn in their own time.

Our attitude towards students also affects their motivation. This is why I always try to be enthusiastic, approachable and helpful. It is nice to see that several students in their feedback (see appendix) mentioned that I was “enthusiastic”, “friendly”, “helpful” and “approachable”. I also do my best not to be condescending when students ask questions or come to see me in my office. This is important so that students who are unsure of their ability do not get demotivated by unnecessary negative comments when they ask questions. Indeed, feedback during the course of the module

affects strongly the second factor listed by Biggs & Tang (2007), which is whether the student thinks he is likely to succeed at a task. If students receive negative feedback like “if you got this wrong, you’re not good enough to be on this course” (“blame the student” approach), they will lose their motivation. I must admit that when I gave the students feedback on their first assessed problems sheet for the EM module, I could have provided more constructive feedback. Many students had failed the vector manipulation part of a question on electric fields, which I thought was basic knowledge that they should have known by that time. I was of course not mean so did not tell them that they were not good enough to be on this course but at the same time I did not tell them how to improve the situation. Instead of just telling them they had not understood the vector aspect of the question, I could have encouraged the students by telling them that many of them still had understood the physics concept that was central to the question and that by revising vector manipulation, they would do much better next time. I could have also repeated to them at that point that if they struggle, they can come and see me or Rob Jeffries, who teaches the mathematics module.

The final aspect about motivation that I think I could improve in the future is to encourage the students at the start of the module. As the example on page 32 in Biggs & Tang (2007) shows, a word of encouragement at the start can change the students’ perspective for the rest of the module. In that example, a lecturer said “Anyone who can’t follow this isn’t fit to be at University” (level 1 teaching approach where students are good or bad and teaching cannot change this) and really knocked the confidence of the student. In the next semester, another lecturer told the student that the outcome depends a lot on the work the students put in, which encourages every students to work hard and think they have a chance of success. At the start of a module, students often feel overwhelmed by the new topic. This was especially the case for the SKE and SASP students who had not studied in years and had to not only get used to my teaching methods but also to the whole programme. The first time I taught the module, I could only observe that they were lost but did nothing about it. The second time I taught the module, I told them that they would probably feel lost and that

I would go as slowly as possible to explain all the new concepts but most of the students would not be encouraged by that news and would not think that they had a better chance of success after what I told them. I now plan to encourage them better by explaining to them that if they felt lost on the first day does not mean that they will fail the module. It rather means that they will need to get used to their environment and that they may need to catch up with certain skills or knowledge that are used in the module that I teach, for example mathematics. I plan to tell the students that the probability of success in their studies depend more on their effort than on their ability. Since I like sports, I will make the parallel with tennis. Rafael Nadal or Roger Federer did not become world number one just because they were talented but mostly because they trained harder than other players. Also by comparing Nadal to Federer, we can see that even though Nadal may be considered to be less talented technically, he compensates by putting in more physical effort.

3.2 Constructive Alignment (CA)

As [Biggs & Tang \(2007\)](#) nicely explain it on page 54: "Constructive alignment is a marriage between a constructivist understanding of the nature of learning and an aligned design for teaching that is designed to lock students into deep learning".

Constructive alignment first of all refocuses teaching on learning. A second important point is that it focuses on how student learn, rather than on what they learn. [Biggs & Tang \(2007\)](#) breaks CA into four steps:

1. "Describe the intended learning outcome (ILOs) in the form of a verb (learning activity), its object (the content) and specify the context and a standard the students are to attain."
2. "Create a learning environment using teaching/learning activities (TLAs) that address that verb and therefore are likely to bring about the intended outcome."
3. "Use assessment tasks (ATs) that also contain that verb, thus enabling you to judge with the help of rubrics if and how well the students

performances meet the criteria.”

4. “Transform these judgements into standard grading criteria.”

It therefore offers a theoretical framework based on students learning to design modules and programmes. More importantly, this theoretical framework is the key to evaluate, assess and improve our teaching methods and activities. This is especially true in my case. I started teaching before joining the TLHEP so I designed my modules based on my own experience and intuition and by looking at what my colleagues had done. From the students achievements and feedback, I could see that my teaching was alright but I had no theoretical framework to explain why some aspects were working and also it was very hard to improve my teaching without a better understanding of how students learn. I can now combine constructive alignment (providing guidance) and reflective practice (providing a method to change and evaluate the change in my teaching methods, see Sect. 2.4) to evaluate and try to improve continuously my teaching practice. In the next three chapters, I present the evaluation and reflection on four important aspects of teaching: design (ILOs) , delivery (TLAs) and assessment (ATs) and feedback. As explain above, I will use constructive alignment as the theoretical framework to do so and also propose action plans in order to improve my teaching practice.

4 Design

Before discussing about the modules that I teach, it is important to present the general context in which my modules are taught since the intended learning outcomes of my modules have to follow the general ILOs of the physics/astrophysics degree. Note that the next section is part of the group work with Dean McLaughlin and Joana Oliveira on discipline specific pedagogic project. The presentation we gave to other TLHEP participants can be found in Sect. ???. Several quotes are thus the same as in the presentation or very similar. I therefore acknowledge contributions from my colleagues for the discipline specific aspects of our teaching and in particular the sections discussing teaching in general at Keele and in physics. I also acknowledge many valuable discussions with both Dean and Joana during the course of the TLHEP and have tried to identify their direct contributions where I use them in the text.

4.1 General context of the physics degree at Keele

The Institute of Physics is the organisation that accredits physics degrees in the UK. This accreditation is very important because it proves that the physics and astrophysics degrees taught at Keele follow the same high standards as those of all the other major universities in the country. The Institute of Physics published a document called “The Physics Degree: Graduate Skills Base and the Core of Physics” (see [IOP, 2009](#)) in conjunction with the Quality Assurance Agency for Higher Education benchmark statement for “physics, astronomy and astrophysics” (see [QAA, 2008](#)). That last document says that “All accredited degree programmes will comply with” these benchmarks. I acknowledge here that Dean McLaughlin searched for these documents and emailed them to me. He also took the lead in checking

the alignment between our teaching methods at Keele and the guidance provided by the IOP and QAA for the presentation although I am doing my own alignment check below. The IOP document details the skills and achievements that graduates should have:

1. “Degree programmes should provide a positive experience of physics and should encourage the student to foster and maintain an intellectual curiosity in the discipline.”
2. “All degree programmes including joint and combined honours must impart a secure knowledge of the fundamental elements of physics as expressed by the Core of Physics. However, the Institute expects that programmes will be taught to a considerably richer curriculum than the Core of Physics and will include advanced material reflecting the specialist interests of the department.”
3. “All programmes should enable students to acquire the skills listed in the Graduate Skills Base. These skills consist of both physics skills and generic (transferable) skills.”
4. “BSc degree programmes must incorporate either project work or a dissertation; integrated Masters programmes must incorporate extended project work. Project objectives are discussed in the section Project Work.”

The IOP document then lists the topics that constitutes the core of physics: “Mathematics for Physicists, Mechanics and Relativity, Quantum Physics, Condensed Matter Physics, Oscillations and Waves, Electromagnetism, Optics, and Thermodynamics and Statistical Physics”. In particular, for Electromagnetism, the following core knowledge has to be learnt:

1. “Electrostatics and magnetostatics”
2. “DC and AC circuit analysis to the level of complex impedance, transients and resonance”
3. “Gauss, Faraday, Ampère, Lenz and Lorentz laws to the level of their vector expression”

4. “Maxwell’s equations and plane electromagnetic wave solution; Poynting vector”
5. “Electromagnetic spectrum”
6. “Polarisation of waves and behaviour at plane interfaces”

Items 1 to 3 are the topics that I have to cover during my module on Electricity and Magnetism.

The IOP document also states that:

“Physics is a hierarchical discipline, therefore, before some of the topics identified below can be treated in adequate depth certain prerequisite material must also be covered.”

We can see from the last sentence that constructivism is at the heart of teaching and learning in physics, although I had not been able to make this explicit connection before starting the TLHEP since I did not know about constructivism. Knowledge is built on previous knowledge and gradually broadened, complexified and generalised in a spiral curriculum, where most modules have a long list of pre-requisites. This provides a clear structure for learning but, at the same time, it imposes the coverage of a long list of topics, which makes it hard to prevent curriculum overload. We cannot skip content in the first year like special relativity otherwise our students will not be able to assimilate advanced topics like general relativity. In other fields of study, where there is less direct dependence between different topics, it is probably easier to decrease coverage in order to give more time to students to engage deeply in a certain topic but in physics, we are forced to cover all the topics that the IOP lists as core physics. This is a real challenge at Keele with the dual honours system, where we cannot count on the other subject to cover the core physics topics, whereas at other Universities, students taking single honours physics will have twice the time to cover the topics as well as mathematics. Dean McLaughlin calculated the fraction of modules taken by core modules and found that the core physics topics add up to 80% of the dual-honours physics credits so there is little room for improvisation. Mathematics is a necessary skill that many of our first years lack confidence in even though there is a specific mathematics module in the first year. In order to improve the situation, Prof Rob Jeffries is proposing to implement

a mentor-mentee system for mathematics between second and first year students. Even though the IOP imposes a list of topics to be covered, the IOP documents also explains that: “The Core of Physics contains a set of headings under which appear topics that should be covered in an accredited physics degree. As such, it should not be read as a syllabus and a traditional arrangement of the curriculum is not a requirement, nor is a traditional teaching approach. It is more appropriate to read the Core of Physics as a set of key concepts that should be familiar to a graduate of an accredited degree programme. This particular arrangement of the material is given as one possible example.” This opens the door for creativity in module and course design, which should be explored.

The QAA document lists (in Sect. 4) the graduate attributes, split into two categories: Physics skills and generic skills. The physics skills are the following:

1. “Students should learn:
 - (a) how to formulate and tackle problems in physics. For example, they should learn how to identify the appropriate physical principles, how to use special and limiting cases and order-of-magnitude estimates to guide their thinking about a problem and how to present the solution, making their assumptions and approximations explicit
 - (b) how to use mathematics to describe the physical world. They should have an understanding of mathematical modelling and of the role of approximation
 - (c) how to plan, execute and report the results of an experiment or investigation. They should be able to use appropriate methods to analyse their data and to evaluate the level of its uncertainty. They should also be able to relate any conclusions they make to current theories of the physics involved.”
2. “They should be able to compare critically the results of model calculations with those from experiment and observation.”

I listed these graduate skills/attributes (which are the intended learning outcomes at the degree/programme level) in details in order to evaluate

how the ILOs of my modules match these ILOs in the next section. Before doing this I would like to make the connections between the above ILOs and levels of understanding (based on Bloom's Taxonomy [Bloom, 1979](#)) described in Sect. [2.1](#). This link is intuitive for most people but I had personally not attempted to compare the two before and I think this is an useful activity for me in order to assess if the intended learning outcomes that I choose promote deep learning approaches and are linked to high levels of understanding.

The first ILO/skill is “how to formulate and tackle problems in physics”. We can first notice that neither this nor any other skill uses verbs like memorising, describing, or even explaining topics that are low level learning activities. Instead, the examples given: “how to identify the appropriate physical principles, how to use special and limiting cases” requires analysis, synthesis and evaluation of the topic learnt. The second skill, “how to use mathematics to describe the physical world” highlights the fact that physics cannot be applied without the use of mathematics. It is closely linked to the application of the topic. The third skill, “how to plan, execute and report the results of an experiment or investigation” deals with practical skills that are necessary for graduates to undertake research within the world of academia or in the industry. I will not talk about this skill in details in this portfolio because I am not involved directly in the teaching of labs but I definitely appreciate the importance of practical skills and part of the assessment of all the modules that I teach include assessment of experimental skills via the marking of lab reports. The final skill, “They should be able to compare critically the results of model calculations with those from experiment and observation.” corresponds to the highest levels of learning, synthesis and evaluation. We can thus confirm from this comparison that ILOs are properly set in order to encourage deep learning approaches and develop high levels of understanding. We also see in the list of ILOs that the focus is on what and especially how (to which learning level) students learn.

I also list here the generic skills since I want refer to them in Sect. [5.6](#):

1. “Problem-solving skills - physics degree programmes involve students in solving problems with well-defined solutions. They will also gain

experience in tackling open-ended problems. Students should develop their ability to formulate problems in precise terms and to identify key issues. They should develop the confidence to try different approaches in order to make progress on challenging problems.”

2. “Investigative skills - students will have opportunities to develop their skills of independent investigation. Students will generally have experience of using textbooks, and other available literature, of searching databases and the Internet, and of interacting with colleagues to derive important information.”
3. “Communication skills - physics and the mathematics used in physics deal with surprising ideas and difficult concepts; good communication is essential. A physics degree should develop a student’s ability to listen carefully, to read demanding texts, and to present complex information in a clear and concise manner.”
4. “Analytical skills - physics helps students learn the need to pay attention to detail and to develop their ability to manipulate precise and intricate ideas, to construct logical arguments and to use technical language correctly.”
5. “ICT skills - during their studies, students will develop their computing and ICT skills in a variety of ways, including their ability to use appropriate software such as programming languages and packages.”
6. “Personal skills - students should develop their ability to work independently, to use their initiative and to organise themselves to meet deadlines. They should gain experience of group work and be able to interact constructively.”

Obviously not every programme level physics or generic ILO can be covered in every module. For example, we cannot expect first year students to undertake self-directed project so the ILOs for each module will address a subset of the full list of programme ILOs. In the next section, I will assess the alignment of my module ILOs with by the programme ILOs by comparing them.

4.2 ILOs at the module level

One of the main findings I made during the micro-teaching event (see Sect. ??) is that I did not understand the concept of ILOs. Obviously, the micro-teaching was at the start of the TLHEP so most of the concepts that I now have learnt were totally foreign to me. At the time of the micro-teaching activity, I felt completely lost so concentrated on my previous experience as a teacher and the delivery of my teaching generally went well. I kept to the assigned time, explained clearly the theory and set up worked example for the students to apply the newly learnt topic by solving a question. This in my opinion helps providing feedback for the students to assess if they really understood what I had explained and also bring variety in the learning activities instead of having a continuous monologue. I will come back to the idea of mixing lecturing and problem classes, which are usually separate in the next chapter. The feedback from my colleagues was also positive although not very precise. But I knew at that point that I did not really understand ILOs as demonstrated by the ILOs that I wrote on my third slide:

- “Define general properties of stars”
- “Explain relations between mass and lifetime of stars”
- “Apply theory to a problem and solve numerical equations”

I also list here the ILOs that are listed under the heading Objectives in the mechanics modules that I teach:

- “Introduce and develop basic concepts of Mechanics by lectures, problem classes and laboratory work. ”
- “Students should have achieved the following objectives at the end of this module:
 - Sufficient understanding of the relevant theoretical and experimental background of Mechanics,
 - General physical significance and applications of Mechanics,

- Able to tackle problems and numerical calculations in connection with mechanics.”

For the EM module, which I started to teach this year, I attempted to improve the ILOs and changed the heading from objectives to ILOs:

- “Understand the theoretical and experimental background of electricity of magnetism,”
- “Appreciate their general physical significance and applications,”
- “Ability to carry out numerical calculations and to solve problems in connection with these topics.”

Before evaluating these ILOs, it is necessary to define clearly what ILOs are supposed to be. For me it is reassuring to know that it is only recently that ILOs have been used and that before the heading “Objectives” was used, even in the previous edition of [Biggs & Tang \(2007\)](#) as they explain on page 70. This means that I am not the only person for whom the concept of ILO is new. The main reason why the term ILO is used instead of objectives is to centre ILOs on what the student is supposed to learn. This is done by having the students as the subject of each sentence describing the ILOs. Therefore ILOs should start as “Students should learn“ as in the physics degree ILOs and “Students should be able to ...”. We can see that the ILOs of the micro-teaching are more focused on what I was going to do as a teacher than on what students would learn. It is easier for the students to know what they are supposed to do if we explain to them what students have to do rather than what the teacher is going to do.

The other main points that ILOs have to address is “What learning activities are involved? What level of understanding are the students to achieve?”, as explained by [Biggs & Tang \(2007\)](#) on page 70. We have already presented the taxonomy to describe the level of learning/understanding developed by [Bloom \(1979\)](#). This taxonomy has been revised by [Anderson & Krathwohl \(2001\)](#) to include the following headings: “remembering, understanding, applying, analysing, evaluating, creating”. The first five levels are very similar to the original ones whereas “creating” is a new level. The other taxonomy often used to design ILOs is the SOLO (Structure of Observed Learning

Outcomes, see [Biggs & Tang, 2007](#), pages 76-82), with the following levels: uni-structural, multi-structural, relational and extended abstract. Both taxonomies are hierarchical but look at activities from a different point of view. I would say that Bloom's taxonomy deals more with types of activities (the nature of the activity) whereas SOLO deals more with cognitive processes.

If I now look back at my ILOs for the EM module, I can immediately see that the levels of understanding that the students are supposed to achieve are vague or missing. Here is a new version of ILOs that I will use in the future:

At the end of the module, the students should be able to

1. State and explain the main laws (e. g. Gauss, Faraday, Ampère) and concepts (e. g. fields, inductance, impedance) in the field of electricity and magnetism.
2. Apply concepts and laws to numerically solve problems using the skills learnt in the mathematics module.
3. Conduct experiments and write lab reports.
4. Compare and contrast the electric and magnetic forces/fields with each other and with the gravitational force/field.
5. Identify the theory relevant to solve a problem and choose the best method if there are several possibilities (Biot-Savart and Ampère's laws).
6. relate the concepts learnt in the module to applications in everyday life.

The students will also develop the following generic skills during this module:

1. Analytical and problem-solving skills, in particular using different approaches to solve problems.
2. Investigating skills, by using textbooks to support their learning.

I tried to list the ILOs in the same order as in Bloom's revised taxonomy. There is no ILO at the "creating" level but this is acceptable in a first year module. I explicitly put the first ILO because I think that memorising is undervalued nowadays. First year students might think they do not need to remember the laws exactly as long as they can use it. I agree that in order to use it correctly, students have to memorise it but I want to make sure that they master the lowest level of understanding to create a solid foundation for the higher levels.

We can now check that the module ILOs are aligned with the programme ILOs (listed in the previous section, Sect. 4.1) by comparing them. Only the fourth physics skill, "They should be able to compare critically the results of model calculations with those from experiment and observation." is missing from my module ILOs but this is acceptable since my module is a first year module. In the past, I had not written generic ILOs alongside the topic specific ILOs since I thought that most of the generic skills are developed throughout the programme rather than in a particular module. As I discussed in Sect. 5.6, I will specifically add the use of textbook in my ILOs so that students realise the importance of developing self-directed learning abilities.

ILOs are also useful for revisions. At the end of my module, a student asked me how he should revise for my module. I thought students already had a revision strategy in place but by observing a revision session of the third year module delivered by Pierre Maxted (see observations notes in appendix Sect. ??), even third year students do not all yet have a good revision strategy in place. By pointing the students to the ILOs and the content the ILOs apply to, the students should have a clearer idea of what is expected of them and should be in a better position to establish their revision strategy.

Now that the ILOs of my module are set, I can move on to teaching/learning activities (TLAs) addressed in the next chapter and assessment tasks (ATs) addressed in Sect. 6 and evaluate if they are all constructively aligned.

5 Delivery

5.1 Teaching and learning activities (TLAs) in physics/astrophysics at Keele

There is a range of teaching/learning activities taking place during a physics degree. Joana Oliveira compiled the list for the discipline specific presentation (see appendix ??). 40% of contact time takes place in lectures and 60% takes place in activities during which students learn actively: tutorials (scripted problems, repetition, practice), problem classes (informal; open-ended applications; critical synthesis; not memorisation) and labs (experimental skills; critical appraisal, comparison to theory; tasks and learning self-directed in part). The number of contact hours are the following:

Year 1: 10 hours per week

Year 2: 9 hours per week

Year 3 - dual honours: 4-6 hours per week plus project

Year 3 - major honours: 8-10 hours per week plus project and dissertation

By comparing these TLAs with the ILOs listed in Sect. 4.1, we can see the alignment between the two. What is also nice to see is that during more than 50% of the contact time students are learning actively, lectures adding up only to 40% of the total contact time. On top of the contact time, students are expected to spend two hours of self-directed learning for each hour of lecture. The TLAs are thus clearly centred on students learning. If I compare this to my studies at the EPFL in Switzerland, I had

a similar balance between lecture time and active learning times but twice the amount of contact time (also with additional hours of self-study time mostly spent writing problems sheets to be handed in every week). So when I came to Nottingham University for my Erasmus exchange, I first felt like I was studying part time. Although I had spent much more time studying in Switzerland, thus learning more content, I had very little time to self-direct my learning activities. On the other hand, at Nottingham University, I had more time to read books and self-direct my learning, which I think is a better balance than the constant directed learning taking place in the system used at EPFL, even in senior years. I come back to this point in Sect. 5.6.

For all the modules that I teach, which are at the first-year level or equivalent, I am in charge of delivering the lectures and organising the problem classes and someone else is in charge of the labs. The relevant mathematics are taught in separate modules. I will therefore concentrate in the rest of this chapter on lectures and problem classes. Before discussing these two activities, I must first make clear my disagreement with Biggs's strong criticism of the lecture/problem class system in chapter 7, most obvious on page 109, where he says: "Why are lectures so ineffective?" (Biggs & Tang, 2007). It is comforting to see that other major textbooks like Fry et al. (2009) do not share Biggs' opinion. Probably Biggs tries to be provocative to challenge tradition and to encourage reflection and improvement of the system in place but I think that the lecture/problem class system is a really good system that should remain the norm in the junior years of the degree. I agree with Biggs that in senior years teaching/learning activities change and students undertake more self-directed activities like projects and dissertations, which is what is done in physics. I also agree with Biggs that student already in the first year should develop the ability of self-directed learning as I explain in Sect. 5.6. However, here is why I think the lecture/problem class system, is a good system for the majority of junior year modules:

- As Biggs explain, the goal of improved teaching is to help Roberts (not intrinsically motivated students) to achieve higher level learning that Susans would do automatically. Getting rid of traditional lecture

would in my opinion not help but instead demotivate Roberts, even if they had a little motivation to start with. Indeed, on page 108, Biggs agrees with Bligh (1972) conclusion that "unsupervised reading is more effective than a lecture". I think only Susans are able to do this. As Biggs discusses in the same chapter, the more senses we use to learn, the better we retain, so if we compare lectures to reading, in lectures students use the following senses: hearing, seeing, writing and in interactive lectures doing (thinking, reflecting) whereas Roberts will probably only read and see in self-directed reading. Assimilation of the new concepts is very hard and Roberts usually do not assimilate new concepts the first time so if they are expected to assimilate new concepts always on their own, they will lose whatever motivation they had at the start and think they are not good enough. On the other hand, lectures help the student carry on learning even though they do not assimilate the topic the first time they learn it and problem classes help them improve the assimilation and accommodation. This point will developed in Sect. 5.4.

- Self-directed learning requires a strong self-discipline that most students do not have, even Susans. If I take my example, even if I am intrinsically interested in many topics, I still lack the self-discipline to learn things on my own. Structured learning activities like lectures and problem classes really help students to get learning started.
- I think that asking students to do the low level learning on their own and use lectures for higher level learning activities is putting the cart before the horse and neglecting the foundations of learning. Yes, students should be engaging in higher level learning activities during lectures and tutorials but laying solid and clear foundations of new topics during lectures is key to achieve successfully higher order levels. I think this point is especially true in science where there are many threshold concepts that students cannot grasp on their own. If students are not taught the theory in lectures, they might as well in this case do a distance learning course. For this discussion, it would be very interesting to compare drop out rates for students enrolled in distance learning

compared to students being on campus.

- Lectures and problem classes are group activities and this is a key to learning. We learn better in groups than by reading a book because we interact during the learning process. On this point, Biggs, me and everyone agrees and this point will be developed in Sect. 5.3 on interactive teaching and active learning.
- Self-directed learning is a great skill to develop but it takes more time than the usual lecture/problem class system and it would take too much time for the students to use this method for all the topics they learn.

Now that I have explained why I think the lecture/problem class system aids learning, in particular for Roberts, I will discuss several aspects of this system. First, I will discuss general good practice for lectures and develop on three specific aspects of the lecture/problem class system: the use of technology (PowerPoint or whiteboard), mixing lectures with problem classes and interactive teaching/active learning.

5.2 Good practice in delivering lectures

In this section, I review good practice in delivering lectures that I have developed or gathered over the years and relate it to the theory of and research on learning. An important aspect already discussed in Sect. 3.1 is to motivate the students to learn by highlighting the relevance and importance of the topic throughout the lecture.

At the start of a lecture, before diving in the topic of the day, there are two things that can be done to improve students learning:

- The first one is to recap what was learnt during the last lecture. This is key to improve the connections between the new knowledge and existing knowledge and therefore strongly helps assimilation ("fit practice [new knowledge] to theory [existing knowledge]" using the definitions by Atherton (2011) introduced in Sect. 2.4).

- The second one is to present a plan or table of contents. This helps the student establish a structure for the new knowledge. [Biggs & Tang \(2007\)](#) calls this “constructing a base of interconnected knowledge”. The plan reveals the horizontal connections between the main topics of the day. This helps the student know which are the main topics that they are learning before going into the small details of each topic. Once the main structure is in place, it is important to establish vertical connections throughout the lecture. I do this using a title for every slide. This tells the students which general topic we are dealing with at any time during the lecture so that they are able to zoom out when trying to fit the new knowledge into their existing knowledge structure. This also helps assimilation because it helps them know which part of their existing knowledge they are supposed to connect the new knowledge to.
- A third information to provide the students with, especially for the first lecture of the module is the ILOs of the module so that the students know the rules of the game. Specific ILOs can also be repeated if they are the main focus of a particular lecture.

During the lecture, varying the pace and the type of activities are important aspects to improve students learning. As the research of [Bligh \(1972\)](#) shows, students can only keep a high level of concentration for about 15 minutes, after which only a change in activity can reset the level of concentration to a high level. I will discuss different types of learning activities in the next sections and how they help maintain concentration and also the learning process. The other aspect is pace. It is important to take sufficient time to clearly explain (sometimes twice) key concepts, in particular threshold concepts ([Cousin, 2006](#)) and compare, contrast and relate them to whatever existing knowledge the students might possess. One such threshold concept in the EM module is phasors. Phasors are a mathematics tool (linked to cos and sin functions) used to describe the time evolution of the current and voltage in AC circuits. The concept is abstract and not intuitive since we are using vectors to follow the time evolution of scalar quantities. I therefore spent more than half of a lecture just on this key concept. I first used a key drawing of phasors as my title slide to signpost

the importance of this key concept. At the start of my lecture, in order to motivate the students to know more about phasors, I told them that if someone could explain to me everything about this diagram, they could leave the room. This is in my opinion much better than just telling them that this is a hard concept and that we have to spend most of the lecture trying to understand it. On top of motivating the students, it helped them access whatever existing knowledge they had on that topic, which helps the assimilation of new knowledge. The students contributed answers for about 15 minutes, which was a bit long. As Dean commented when observing me, I tend to spend too much time asking the students to recap what we have learnt in the previous lecture. I agree and will from now on make the recap part of the lecture shorter by providing myself some of the previous knowledge instead of waiting for the students to provide the full list. At the end of this interactive start, students had contributed all but one key explanation about this threshold concept so they still had to attend the rest of the lecture but their mood had been very good during that part of the lecture. Since phasors is a difficult and important concept, I did not mind reviewing again all the explanations the students had provided at the start and I could also correct misconceptions about the concepts along the way. On the other hand, I usually spend less time on other parts of the lecture that I think are easier to understand or that students can look at after the lecture. This is probably something that I have to better explain to the students because many of them complain that the pace is too fast. For example, these SASP students thought the worst aspect was the pace:

- “Cover too much for one session. Unfortunately it is the course I think”
- “Sometimes, pace is a little quick for me! Too much info in one two-hour slot!”

It is also the danger of using PowerPoint, which I discuss in more details in Sect. 5.5.

Connecting the new knowledge to the existing one is a central point of constructivism. When introducing new concepts, I spend time comparing and contrasting new concepts to old ones. Comparing helps assimilation and contrasting helps accommodation of existing schemata in order to cre-

ate new schemata for new concepts. Using worked-example straight after the introduction of the theory is very useful so that students can learn how new concepts are applied. I discuss this point further in Sect. 5.4. Near the end of the lecture, it is sometimes possible to relate the different parts of the lecture together in more complex worked-examples or applications, which helps student relate the different parts of the lecture. At the module level, near the end of the module, relating the different parts of the module together by teaching more complex concepts and applications involving several previously covered concepts of the module is also a good way of aiding students structuring of the new concepts as parts of a "whole" system. For example, near the end of my module, I organised a quiz where students had to relate what they had learnt up to now on AC circuits with the similar concepts they had learn in Dean McLaughlin's module "oscillations and waves", thus helping the students seeing different modules as related parts of the whole physics knowledge.

Finally, it helps to select a reasonable amount of material to teach within a session so that there is time at the end to summarise what has been taught during the lecture. The summary helps reiterate the main concepts taught during the lecture and consolidates the main structure of the content. Research by Bligh (1972) has shown that this also considerably helps the retention of the material learnt during the session. Up to now, I always read the lecture summary to the students at the end of the lecture but Biggs suggests to ask the students to do the reviewing. I think I have to find the right balance between time spent and students active participation for both the recap at the start of the lecture and the summary at the end of it. Another good idea is to ask at the end of a lecture what students found most difficult or confusing to understand or do. This should expose threshold concepts and also help the students find out what they need to study in more detail before the next lecture. This is a technique that Eric Mazur, physics teacher at Harvard, used as preparation activity (as reported in Biggs & Tang, 2007, page 111). Most students at Harvard are probably Susans who are able to learn most topics by reading the relevant textbook but I think it is more reasonable to ask Roberts to do this type of activity after the lecture, as discussed in my opinion on the lecture/problem class

system above. I also think that asking students to read textbooks after the class, which is one of my new generic ILOs discussed in Sect. 5.6, is more reasonable for Roberts, who would easily be discouraged if they had to learn new concepts in textbooks on their own for the first time.

5.3 Interactive teaching and active learning

As most people would agree, one-way delivery of lectures is not the best way for students to learn. As discussed above, this is in part due to the fact that most people can only concentrate on the same task for about 15 minutes (Bligh, 1972). The other main aspect is that only deep learner are actively learning without being asked to do so. The art of lecturing is to encourage and get most students, especially the Roberts, to use deep learning approaches like Susans (see Biggs & Tang, 2007, page 10). How do we achieve this? The simplest activity that achieve this purpose is questioning the students. Fry et al. (2009) calls it “interrogating practice“. The idea is to ask the students the questions that Susans would ask herself or that the lecturer would ask her/himself when preparing the lecture material. The difficulty is to get every student, not just Susans to provide an answer to these questions. There are different ways to maximise the fraction of students that actually answer the questions, which I have discussed with Rob Jeffries and Dean McLaughlin. One can ask the students to raise their hands, although some students might copy the rest of the class. One can use individual electronic voting systems, which are very good but not so easy to implement and necessitate the correct hardware and software equipment. A possibility is to give individual whiteboards to students so that each of them show their answer to the lecturer. This is what Rob Jeffries used in the maths lecture that I observed and I think it is a great method, which I will try in the future. One could also ask the students to write down the answer on their own notepads, which mean they can keep a record of their answer. I quite like this method. Another option is to pick the student or group of students who is going to provide the answer. There is a limit to picking students to answer questions, especially in physics, where many students have a pre-disposition or a disability (aspergers, ..., see full list

compiled by Joana in the discipline specific presentation in appendix ??) that makes them uncomfortable to speak in front of the whole class. The goal is indeed not to create a climate of fear or anxiety in the students but at the same time, a small positive pressure has to be felt by the students in order to encourage them to actively learn during lectures. The pressure is usually felt at the start of the semester when students are not used to actively participate. The pressure can be diminished by asking a small group of students to provide the answer or by asking students to work in small group before they provide an answer. The pressure can also be reduced by giving advance warning to the student that will have to answer the question, which is what Rob Jeffries did during the lecture that I observed. Once the students are used to the questioning, the pressure usually becomes a motivation. The students are usually more affected by the feedback they receive after providing an answer than by the pressure they experience when asked to provide an answer. If the lecturer laughs at their answers and provides unnecessary negative feedback, the student will unlikely want to provide another answer. Biggs & Tang (2007, on page 98), talking about handling students error in student activity provides the following guidance that is also valid for handling students answers:

“Using errors constructively thus involves two challenges:

- requiring students to expose their erroneous thinking without the risk of ridicule, loss of face or low grades.
- correcting them nicely so that they feel positive about being corrected and not ashamed or resentful.”

Here is an extract from the analysis done by Cavanagh (2011) of students feedback about active participation in class (which is discussed further in the next section):

“Some students noted that, even though they were apprehensive at first about sharing with peers in the lectorials, they persevered because of the benefits which they gained as a result of their participation: ‘Sometimes just to talk to the person next to me, though uncomfortable at first, allowed me to lift the level of my own learning.’ Student-to-student interactions, either with a partner or in small groups, were found by many to ‘really help illus-

trate a point and make it much easier to understand', principally because they were 'discovering something for ourselves' in a supportive learning environment. The concepts presented in the lectorials were reinforced through constant repetition during the small-group and whole-class discussions, and hearing from peers in their own words gave the ideas additional credence."

The main point I want to stress from this analysis is that by asking students questions (however small and simple) allows the students to "discovering something for ourselves" and that this is a very motivating experience for them. Note that the level of learning required to answer the question is also important and should be chosen appropriately depending on the goal of the questioning. There are different use for questioning as described in [Cavanagh \(2011\)](#) study:

"Sometimes the lecturer presented his ideas before students were given a related task which allowed them to think more deeply about what they had heard, and on other occasions the cooperative learning activity was used as a prelude to the lecture-style delivery so that students had already started thinking about the issue and were more receptive to the lecture content which followed."

The two uses of active learning activities (also applicable to questions) listed in this paragraph are linked to inductive (asking the students to contribute knowledge before teaching the topic) and deductive (teaching the topic first and then ask the students questions or ask them to apply what they have learnt in problem class) methods of teaching. As I said in Sect. [1.1.2](#), I thought only deductive methods were useful in science because I thought inductive methods were confusing at best. Now I have realised that both are valid method but they should be used for a different purpose. An inductive method should be used to aid assimilation of new concepts by activating students existing knowledge about it (for e. g. threshold concepts of phasors discussed in the previous section). A deductive method should be used to aid accommodation of new concepts by for example asking the students to compare and contrast the electric force to the gravitational force or asking the students to explain the difference between AC and DC circuits after having introduced these new concepts.

How do questioning and other active learning activities (I will discuss

mixing problem classes and lecture next) help student learn? It first provides immediate feedback to the students and the teacher about how well they have understood the topic. It can for example highlights misconceptions that are due to imperfectly adapting an existing concepts to the new situation. Even though the lecturer could highlight common misconceptions or compare and contrast new concepts to existing ones, it makes it more real and easier to remember for the students if they themselves provide this information. The more *relevant* activity the students do, the better they learn as presented in Table 6.1 on page 96 of Biggs & Tang (2007) (source attributed to William Glasser): students learn more and more as actions change in the following order: “(1) read, (2) hear, (3) see, (4) see and hear, (5) talk over with others, (6) use and do in real life, (7) teach someone else“. This agrees with the saying of my high school teacher: “we listen and forget, we write and remember, we do and understand“. This is why tutorials, which are called problem classes in sciences are such an important part of the teaching and learning activities.

Interactive teaching is trying to use small group activity into large group teaching context. Even though it is harder to manage as the class gets larger, many teachers are able to teach interactively to large group. Personally, I found it easy to use interactive teaching in groups up to 30 but find it a bit more difficult with groups larger than 50 but it might just mean I need to get used to manage bigger classes. Biggs & Tang (2007) reports studies saying interactive teaching works with classes up to 80 on page 112. I focused in this section mostly on questioning but there are other activities that could be used in lectures (see Biggs & Tang, 2007, chapter 7 for several ideas). I think however that some group activities involving a lot of discussion (buzz group) are not suitable for sciences as presented in our discipline specific presentation (see appendix ??). The other main learning activity I discuss next is problem classes.

5.4 Lectorials

The three main teaching/learning activities in physics are lectures (discussed above), problem classes and labs. Labs are designed for students

to learn experimental skills but also to compare theory to real data, analyse data and thus students learn to evaluate theories in the process. In problem classes (science tutorials) students solve problems using the theory they have learnt in the lecture. During problem classes, students get help from PhD students and a staff member, sometimes the lecturer in charge of the module. In the problem class they therefore apply the new concepts they have learnt and deepen their understanding of the topic while developing valuable generic problem solving skills. For my mechanics modules for the professional programmes SASP and SKE for future or existing school teachers, I am free to organise as I want both the lecturing and problem classes activities in 2-hour slots. The labs are organised separately so I will not discuss them here. Instead of planning a 1-hour lecture followed by a 1-hour problem class session, I decided to mix both in one 2-hour session. Michael Cavanagh, at Macquarie University, Australia, has done the same experiment and published his analysis of the experiment early this year, see [Cavanagh \(2011\)](#). He called his mixed sessions "Lectorials", since the sessions are a mix of lecture and tutorials. In this section, I am going to discuss both my own and Cavanagh's experiments of mixing lectures and problem classes (discussion that I presented at the discipline specific presentations, see appendix ??).

The idea of mixing problem classes and lecture is very old. It is actually what I and I guess most people experienced when learning physics at school and at A-levels so it is a surprise that there is not more research published on this topic as noted by [Cavanagh \(2011\)](#). Asking students to solve problems during lectures is an obvious option to bring variety to learning activities in lectures, thus "Maintaining interest and attention", as [Cavanagh \(2011\)](#) named one of the sections in his paper. Both his students and mine (although I did not ask this specific question in the feedback form) agreed that the mix helps maintain interest and attention confirming the intended purpose. In a similar study done by [Huxham \(2005\)](#), the interactive nature of the lecture was the most popular aspect according to the students' survey. This shows that at least students' motivation is definitely increased by this type of activity and it is therefore worth considering this activity in teaching. The other main question is "whether the lectorials help stu-

dents better understand the content taught in the sessions”. I asked the SASP students this question and their answers are: “yes / yes definitely / Yes, help to consolidate and test learning / Yes / A little” and the SKE students answers are: “Yes but needs to go through them slower / Problem solving sessions could have been slower / Yes, very important to aiding understanding” (see SASP and SKE student feedback in appendix ?? and ?? at the end of questionnaire). This shows a general positive answer to the question. Cavanagh (2011) also obtained mostly very positive response from his students: “All but three of the 94 students who had attended at least 11 of the 13 lectorials agreed”. However, the study of Huxham (2005), by comparing the results of assessed tests of two groups, with only one following lectorials type sessions, showed that the students following the lectorial-type session performed only slightly better than the other group. Other studies discussed in Cavanagh (2011) after Huxham’s study do not find any positive impact of the lectorials on students learning:

“For example, Huxham (2005) noted a generally weak positive influence of interactive learning tasks in lectures on science students’ recall and learning. Vreven and McFadden (2007) found that students undertaking a three-week general psychology course derived no additional benefit from cooperative learning activities in lectures because the course was time-compressed with a large number of students, while van Dijk et al. (1999) concluded that the skills and preferences of the lecturer were more crucial than the provision of cooperative learning activities.”

I also asked my SASP/SKE students the following question: “Would you prefer two separate 1h theory (lecture) and problem class sessions?”. The reply I received from the SKE students are: “Yes but still do examples in theory / Yes / No, prefer problems interspersed with theory” and from the SASP students: “No, I think it works well addressing problems before moving on. / Yes, opportunity to get stuck into comments and more at my own pace and take rest home having made a good start. / Happy with current structure / Yes, going through problems with help would be good / Yes” (see appendix ?? and ?? at the end of questionnaire). From these responses, we can see that at least some students would like to have a separate problem class session. It might just mean that they would like

more contact time in general as one student comment implies: “Yes but still do examples in theory”.

Reflecting on the my own experience of lectorials and reading students comments, I noticed important issues and ways to improve. Some students commented that the problem solving sessions are rushed so that they do not feel they have the time to get stuck in it. One suggestion that Stephen Bostock made to me after observing one of these sessions is that I could split the problem solving sessions into two parts. First a “find the strategy” part that hopefully most students can finish and discuss pretty quickly and then a second part during which the students attempt to solve the problem. Even though most would not have the time to finish the second part, at least everyone would have done the first part, which is the most important part to solve problems.

I also noticed that during the lectorials, a few good students finished the problems quickly and then had to wait until the rest of the class had finish and sometimes got bored. This was especially the case for the SKE and SASP modules where students come from very different backgrounds and start my module (the first module in the programme) with very different skills in mathematics as well as mechanics. I tried to improve the situation by breaking the problems into a basic section and an advanced section. Doing the basic section covered the concept that I wanted the students to learn and the advanced one would relate it to the other concepts already covered. I would then resume the teaching part once everyone had finished the basic section and usually the best students did not get bored with this method.

We see that introducing problem solving activities is not necessarily an activity that enhances learning and great care has to be taken to design relevant activities that keep the better and weaker students busy learning and at the same time not too long so that sufficient material can be covered during a lecture. Most studies show that including problem solving activities (or questioning as discussed above) are in any case a good way to get the students doing the learning and definitely helps with motivation and concentration and so it is worth using them in the lecture context.

5.5 PowerPoint or white board or both? What else?

I will now focus on the use of technology in the classroom. The two main types of “technology” used in the classroom are white boards and PowerPoint style presentations. Writing on the white board is slow enough for students to take notes comfortably while still having time to think and process what they are being taught. However, hand writing can be hard to read and it is not easy to draw nice graphs or pictures on the board and I use a lot of graphs in my lectures. On the other hand, using PowerPoint (and handing out the slides to the students or putting them on a website for them to download) means that they do not need to write down everything so can focus on listening and digesting the materials presented. It also allows the teacher to quickly go through slides and ask the students to read them later although this is usually a dangerous trap. Indeed, it is very easy to go too fast through the material when using PowerPoint. Even though I know this common trap, I still fell into it according to the EM feedback forms (see appendix ??, worst aspects, response 2: “lectures were sometimes rushed” or response 20: “The overly large amount of PowerPoint slides presented by the lecturer”). This was the first time I taught this module so I wanted to make sure the students (and I) had all the information on the slides. In the future, I will trim down the information on the slides and the number of slides to improve the situation. I will also prepare two versions of the slides, one to use during the lecture and one more complete for the students to use outside the lecture as I did for the SKE and SASP students. I also will try to improve the balance between the use of both technologies, which took me a few years to achieve for the SASP and SKE mechanics modules. Indeed, the first time I taught the SKE programme, I had all the information on my slides, even the answers to the problems that the students had to solve during the class and I handed out a print out of slides at the start of the session. Students could then easily drift into a surface learning approach by looking at the answers on their printout instead of trying to attempt to solve the questions on their own. The following year I easily prevented this surface approach by blanking out the answers to the

problems in the slides I was handing out to the students. I still provided the students with a different version of the lecture slides including model answers so that the students would be able to evaluate their learning by comparing their answers to mine.

Also, with my confidence growing when teaching a topic year after year, I do not feel the need to write as much on the slides and spend more time talking, which means that the slides get less and less crowded year after year.

One technique that I want to explore in the future is using "Note taking should be used as a teaching/learning activity, in other words, not as a horrible inefficient recording device." as Biggs & Tang (2007, explain on page 114). For this purpose, I would like to experiment using interactive white boards. This way, the slides I would use during the lecture would be less full and I could write on the slides the most important information. Students would then mostly write the main concepts during lectures and it could help them learn these better since by writing it down, they have to pay attention to what they are writing.

Another technical improvement That I will try to push for in order to have more space to write on in lecture rooms is vertically sliding boards (see for example: <http://www.usmarkerboard.com/Vertical-Sliding-Boards/42645>). This is not a new idea and most lecture room still equipped with black boards have such technology in place. The main advantage of having more surface available is that we keep more content written on the board and thus easily refer to it later in the lecture.

With Internet, there are now many new technologies available to diversify modes of delivery. I already discussed the use of animations in Sect. 3.1 on motivation. In that section, I also mention screencasts, which are popular nowadays and I plan to create such screencasts to explain key concepts.

5.6 Handouts and textbooks

This year was the first time I taught the EM module and it was the first module I taught to undergraduates. Before the start of the module, I discussed with other colleagues and asked them what teaching material they provided

their students with. In particular, I was interested in what they thought about lecture notes and the use of textbooks in their module. The answers were very varied. Some of my colleagues do not have lecture notes separate from their lecture slides but instead use textbooks. Dean McLaughlin and others had to write lecture notes because they had not found a textbook suitable for their module. Other colleagues only expect their students to read the lecture notes they provided the students with and do not necessarily expect their students to read textbooks. In my own experience as a learner, I experienced both situations. In Switzerland, many modules had comprehensive notes so I did not need to use textbooks, although the notes were as long as textbooks. When I did an Erasmus exchange at Nottingham University in my third year, the situation was the opposite. I only had what was written or said in the lecture and was given a chapter of a textbook to read every week for each module. Before my Erasmus exchange, I enjoyed having lecture notes because it was easy to know what I needed to learn since it was all contained in the lecture notes. However, during my Erasmus exchange, I enjoyed even more using textbooks because I could learn more about the topic than just with lecture notes and it provided me with another point of view on the topic. Although I had not attempted to search the exact benefits for learners, I decided that I would encourage my students to use a reference textbook for my module instead of lecture notes. For that purpose, I prepared my lecture material in the following way:

- I used the same figures and variable names as in the textbook so that it would be easy for the students to read the textbook after my lectures.
- For each individual lecture, I told the students and wrote in the lecture slides which chapters in the textbook corresponds to the topic covered that week.
- I provided comprehensive lecture slides but no lecture notes so that the students would not have to read both lecture notes and as well as a chapter of a book every week.

When the students feedback ?? forms came back to me, several students liked my lecture slides. They found them nicely prepared and clear, which

was encouraging. However, many students complained about the absence of lecture notes (e. g. EM feedback, worst aspect, response 19, “No lecture notes aside from the slide shows. Slide shows convoluted and hard to understand”). One student (worst aspect, response 13) mentioned the example of the lecture notes provided by my colleague Dean McLaughlin: “Better lecture notes needed - (in more depth with improved explanations. PowerPoint slides poor (see Dean’s for example, laid out notes and summary sheets).”. Since student learning is the goal, I have to understand why the students complained and how I should improve the situation. On the one hand, providing the students with comprehensive lecture notes helps the students learning by providing all the material they have to learn in one place, thus providing scaffolding for their learning. Lev Vygotsky’s theory (1978) of “zone of proximal development“ (ZPD) could be applied here in the sense that the more material we provide the students with, the faster and easier they will learn. Along the same line, some students have also asked me to provide them with a summary of formulae to learn for the exam. I could then just write new lecture notes for my students but I know that as a student, I enjoyed even more using textbooks than lecture notes during my Erasmus year. After having thought about it in the context of intended learning outcomes, the reason why I enjoyed it more was that the knowledge presented in the book was deeper and I had learnt a new skill. I was able to learn from a book rather than from well prepared, self contained lecture notes. I knew from that point that I could learn using textbooks, which is a key generic skill for self-directed studies and continuous professional development. So instead of just preparing lecture notes that would make student learning easier, I think it is more important for them to develop the skill of learning from a textbook. I had realised that learning with the help of a textbook had from the start been one of my intended learning outcomes (ILOs) although I had not explained it clearly to the students. I had not even listed it as an ILO. So for next year, I will add : “at the end of the module, students will be able to use a textbook to support their learning” on my ILOs list (improved list of ILOs given in Sect. 4.2) because I believe it is a excellent generic skill to develop for the students. This skill is actually part of the generic skills in the QAA benchmark document under

“investigative skills” (see Sect. 4.1). My module only intends to initiate the students to self-directed learning and this skill will be further developed in the second and third years but I think that if I explain this clearly to the students and help and encourage them to read a book, students will have learnt (or improved on) a valuable additional skill during my module and they would have developed as a learner.

6 Formative and summative assessments

There are two types of assessments, formative (assessed work during semester returned to students with feedback) and summative (e. g. exams to provide grades) although, as I discuss at the end of this chapter, even summative assessment can also become formative if feedback is provided to the students. As I did for the teaching/learning activities (TLAs), I first look at the assessment tasks throughout the physics/astrophysics degree at Keele.

6.1 Assessment in physics/astrophysics at Keele

As for TLAs, the assessment tasks (ATs) are very varied (list compiled by Joana Oliveira for the discipline specific presentation, see appendix ??):

- time-constrained examinations (all years)
- open-book tests (maths tests year 1)
- problem-based assignments (all years)
- problems classes (attendance AND engagement, years 1, 2)
- laboratory books and reports (years 1, 2)
- individual project reports (year 3)
- team project reports (year 3)
- oral/poster/seminar presentations (year 3)
- project artifacts such as programs or circuits (year 3)
- viva voce interviews (possible in year 3)

- essays (some modules)

By comparing this list to the programme level ILOs listed in Sect. 4.1, we can see that there are not only varied but especially well aligned with the ILOs. For example generic communication skills are assessed in oral, poster and seminar presentations. The weight of the different ATs is also well balanced. In first and second years, module marks are usually composed of the exam mark (60%), assessed problems sheets submitted during the semester (20%) and laboratory work and reports (20%). In the third year, some modules marks are more heavily reliant on the exam (80%) but there are other modules that do not include any time-constrained exams (e. g. dissertation and project modules). This assessment diversity is in line with the QAA (2008) physics benchmark statement:

”Time-constrained work has its place in testing the student’s capacity to organise work, as well as to think and to communicate under pressure. Such assessments should be augmented by others, such as presentations and project reports, which allow students to demonstrate what they can achieve with less severe time constraints. Skills such as project planning and execution, research skills, application of ICT and report writing, may be best assessed in this way.“

I will now focus on the EM module, which is a first year module. The weight in percentages of the different assessment modes are the following:

- Examination: 40
- Mathematics tests: 20
- Assessed problems sheets: 10
- Laboratory & reports: 20
- Attendance & engagement at problem classes: 10

I repeat here the EM module ILOs in order to compare them with my assessment tasks:

“At the end of the module, the students should be able to

1. State and explain the main laws (e. g. Gauss, Faraday, Ampère) and concepts (e. g. fields, inductance, impedance) in the field of electricity and magnetism.

2. Apply concepts and laws to numerically solve problems using the skills learnt in the mathematics module.
3. Conduct experiments and write lab reports.
4. Compare and contrast the electric and magnetic forces/fields with each other and with the gravitational force/field.
5. Identify the theory relevant to solve a problem and choose the best method if there are several possibilities (Biot-Savart and Ampère's laws).
6. relate the concepts learnt in the module to applications in everyday life.

The students will also develop the following generic skills during this module:

1. Analytical and problem-solving skills, in particular using different approaches to solve problems.
2. Investigating skills, by using textbooks to support their learning.”

There are some obvious match like for the laboratory ILOs and assessment or the mathematics ILO and tests. The examination, assessed problems sheets, and attendance & engagement at problem classes assess all the other ILOs as I will show below. The examination is the main summative assessment method and I will come back to it in Sect. 6.3. The problem classes and assessed problems sheets are both tasks that provide formative feedback to students and I will discuss them now.

6.2 Formative assessment

Formative assessment and the accompanying feedback play an essential role in students learning as Ramsden (2003) says on page 187: “It is impossible to overstate the role of effective comments on students’ progress in any discussion of effective teaching and assessment.”. As discussed in the previous chapter, feedback can be provided to the students during lecture via

interactive teaching methods like questioning but this is not sufficient. The first formative assessment task is problem classes. During problem classes, students engage fully with the topic and develop a deeper understanding by mainly applying the new concepts they learnt in the lecture to solve problems. They also develop the generic skill of problem solving. Problem classes are very interactive and students are able to ask questions about the problems as well as the theory that they are learning. This is thus clearly an activity centred on students learning. The level of the problems to be solved in the class is quite basic since the goal is to help the students understand the theory and apply individual concepts but some questions are also designed to help them compare, contrast and relate different concepts. The attendance and especially engagement is taken into account in the final module mark. Although this can be seen as being too strict and not trusting the students, it gives them a little extra incentive to do the higher level learning. Since only 10% of the full module mark is attributed to this component, it does not penalise students who prefer to work on their own. Note also that two absences are allowed and that absences justified by extenuated circumstances are not counted as an absence.

The other form of formative assessment is assessed problems sheets. There are three sets of assessed problems sheets per module that the students have to solve in their own time and hand in. Even though they have to write their own answer (collusion is penalised), most students are discussing these questions with their peer and this group interaction is welcome. The level of the assessed problems sheets is similar (slightly higher in my case) than the level of exam questions. This is supposed to help the students find out if they have reached the ILOs for the topics covered so far. This is why some of the students complain that the level is too high (e. g. response 15 for worst aspect of the EM module questionnaire: “How difficult the problems sheets were.”, see appendix ??). In the future, I have to better explain that this is as difficult as it is going to get and solving harder problems during the semester better prepares them for the exam. Also students have more time to answers the questions and can use any resources, which is not the case during the exam. On the other hand, it might better motivate the students and build their confidence if the assessed problems sheet questions

were easier. I could try to simplify them next time and see how well they do in the exam.

As we have discussed in this section, students are regularly assessed during their learning but it is important to accompany this assessment with constructive feedback. As discussed in Sect. 3.1 on motivation, if results from these assessments are accompanied by negative statements like: “if you have got this wrong, you are really not good enough”, it will have negative impact on their learning. Personally, I have to remember not to make condescending comments on the mathematics skills of my students and instead encourage them to review the concepts that they struggled with. The feedback must also be timely in order to be most effective. To this end, we have abolished extensions for submission of assessed problems sheets in our department so that we can provide model answers on the same day as the submission deadline. This way, students receive some feedback while the questions are still in their head and they are curious to know how well they have done. Instead of extensions, students are asked to submit another assessed task in order not to penalise them for accidents happening the day of the submission deadline.

In the last two lectures of my module, I organised revision sessions so that the students can obtain clarification about the theory and we also looked at questions from the exam paper of the previous year. For the revision session, I asked the students to attempt the questions of the previous exam paper and warned the students that if they do not ask me any questions during that session, I will just stop the session. This was to encourage them to take responsibility for their learning. Attempting previous exam questions also helps reduce the stress that students face at exam by getting them used to what is expected of them during the exam. In order to give everyone a chance to ask questions, I asked students row by row to ask a question in order to avoid having just a handful of students asking questions. Most students actively took part in the revision sessions and actually asked more questions on the theory than I had expected, which is a sign that they tried to really understand the theory.

6.3 Summative assessment

The final task of the module and the last topic of this portfolio is the actual exam paper. As explained above, it only represents 40% of the total module mark for the EM module, highlighting the diversity of assessment tasks used in our department. I will mostly concentrate here on the EM exam paper since the SASP and SKE mechanics exam papers have a similar structure. The exam paper is designed to assess most of the module ILOs repeated here for convenience:

1. “State and explain the main laws (e. g. Gauss, Faraday, Ampère) and concepts (e. g. fields, inductance, impedance) in the field of electricity and magnetism.”
2. “Apply concepts and laws to numerically solve problems using the skills learnt in the mathematics module.”
3. “Compare and contrast the electric and magnetic forces/fields with each other and with the gravitational force/field.”
4. “Identify the theory relevant to solve a problem and choose the best method if there are several possibilities (Biot-Savart and Ampère’s laws).”
5. “Relate the concepts learnt in the module to applications in everyday life.”
6. “Analytical and problem-solving skills, in particular using different approaches to solve problems.”

I will look at a sample of questions to check how well this assessment task is aligned with the ILOs there are supposed to assess. It is crucial for the assessments to be aligned with the ILOs since, as Biggs & Tang (2007) explains on page 169, “students learn what they *think* they will be tested on”. This means that the students will achieve the intended learning outcomes chosen by the teacher *only if* the assessment is aligned with the ILOs. The EM exam paper is structured in three parts. First, there are 16 multiple choice questions (MCQ), each worth one point. Then there

are 8 short questions, each worth 3 points. Finally, there are two sections, in which students answer one out of two questions. Each of the last two sections is worth 30 points. Thus the total of points is 100.

The first section using an MCQ format is more likely to assess lower level learning since the answers are convergent (only one possible correct answer) by design. Nevertheless, we can still ask questions in a way that not only test memory (ILO 1 above) but also test application and relate the topic to everyday life (ILO 6), as for example, in this question that I asked in my exam paper:

- Question: “A transmission line carries a current of 10^3 A from west to east. The earth’s magnetic field is horizontal, points to the north, and has a magnitude of 0.5×10^{-4} T. What is the force on one meter of the wire?”
- Possible answers: “0.05 N, 0.2 N, 50 N, and 500 N”

The second section is composed of short questions that can mostly address only one ILO. The difficulty of the question thus varies with which ILO is to be assessed. Some questions in that section are relatively easy, testing that students are able to apply the theory they have learnt (ILO 2). In other questions, I challenge the students beyond what they have been taught in the class. For example, in one question, I asked them to: “Sketch two different circuit diagrams for a high-pass filter.”, I have showed them one high-pass filter circuit during a lecture but in order to find the another one, they really have to understand the concept of impedance, which is a key concept in AC circuits, and apply it in a totally new situation (ILO 5). I attributed only 1 out of 3 points to the first sketch, which students could have remembered and the other 2 points for the second drawing, requiring a higher level understanding of the topic.

In the last section, longer questions assess multiple learning levels and ILOs. Let us look for example at the last question on AC circuit of the exam paper:

”Question D2

1. Explain what is meant by *resonance* in an AC-circuit and give an example application. [10]

2. A series LCR circuit is connected to a variable frequency sinusoidal signal generator with a peak voltage of 100 V, $L = 3 \text{ mH}$, and $R = 20 \Omega$.
 - (a) The resonance frequency is 1000 Hz. Determine the capacitance of the capacitor. [5]
 - (b) Calculate the amplitude of the current at resonance. [5]
 - (c) Calculate the current flowing through the circuit when the frequency is 5 times the resonance frequency. [7]
 - (d) Determine the phase angle and whether the voltage is leading or lagging the current [3]

End“

The first part of that question assesses ILOs 1 (“state and explain”) and 6 (“relate to the real world” assessed by providing an “example application”). In the second part, we test ILO 5 (“Identify the theory relevant”) and ILO 2 (“Apply concepts and laws to numerically solve problems using the skills learnt in the mathematics module.”).

By having a variety of questions assessing the full list of ILOs, we can ensure that a student cannot obtain a good grade if they have not achieved all ILOs. The pass mark in the current system is so low (40%) that it is unfortunately quite easy for students to pass a module even if they have only achieve a few ILOs. In the Swiss system, the pass mark is 60%, which ensures that students have achieved a majority of ILOs in order to pass a exam. It is beyond the scope of this portfolio to discuss this point further.

6.3.1 Feedback on exam papers

Examination papers are summative assessments and the only information students receive concerning how well they did is one number, the grade. The national student union started a campaign last year in order for students to receive more feedback than just a grade on exam papers. Keele gave students stickers that they could stick on their exam papers if they wanted to have feedback on their paper. After some discussions at our regular teaching meetings, the following feedback system was put into place for the last winter semester for physics and astrophysics students:

- First year students could collect their exam papers after they had been marked and the exam papers had sufficient feedback written on them so that the students knew what they did wrong and how they could obtain the right answer. All the students were told by email (not only the ones who had used the feedback sticker) that they could collect their exam paper from the school office, which was easy for the students since they were on campus everyday at the time (start of spring semester). It turned out that only about 35% of students came to collect their exam papers. The grades of the students who came to collect their exam paper did not show any particular trend towards low or high grades. Some students came to the school office for other reasons but did not ask for their exam papers and some students who had used the feedback sticker did not collect their paper.
- Second and Third year exam papers have to be kept by the department in case there is an appeal. So instead of giving the papers back to the students, students had the opportunity to come to look at their exam papers and the solutions during two sessions. The sessions were chosen so that most students would be free and most probably on campus. Two staff members were also at the sessions to answer students questions. Although the department did not record attendance at the sessions, the general impression was that not so many students had come to see their exam papers.

I am sure that some students appreciated the feedback but the fraction of students that collected exam paper is much lower than one could have expected during the campaign. How can we explain this low fraction, even though it was easy for students to collect their papers from the school office (year 1) or come during the sessions (year 2 and 3)? I think the main reason is that the module was over so whatever feedback the students received could not affect their results anymore. This would be another argument supporting the importance of results and assessment in the mind of students, which I already discussed at the start of Sect. 6.3 and in Biggs & Tang (2007, page 169 and references therein). This then reinforces the importance of aligning assessment tasks with ILOs if we want the students to achieve the ILOs.

7 Conclusions and Outlook

Before starting the TLHEP, I knew very little about learning theory as discussed in the introduction. I only knew that some teaching methods worked better but could not explain why. Neither could I really assess or improve my teaching. This course thus represented a steep learning curve for me and I learnt a lot in the process. Most importantly, I have now learnt a theoretical framework, constructive alignment (based on the constructivist theory of learning), that allows me to explain why some of my teaching methods worked. For example, I always thought that using a title for every slide was useful but could not explain why. Now I can justify it by explaining that it provides vertical connections that students (or any person attending a presentation) can use to structure new concepts into their existing knowledge base because the title tells the students, which main topics the new concepts belongs to. In a similar way, plan (or table of contents) provide horizontal connections between the main topics that are taught during a lecture.

I also discovered many aspects of my teaching that can be improved through reflective practice, for which Kolb's cycle 2.4 provides the theoretical framework. The main aspect that needed improvement is the concept of intended learning outcomes (ILOs), as discussed in chapter 4. Defining ILOs in the constructive alignment framework focuses teaching on student learning. It also helps the students know what is expected of them during the module since students are the subject of the sentences used to describe ILOs. Using constructive alignment, I then checked that my teaching/learning activities and assessment tasks were aligned with the ILOs of my module as well as those of the whole physics degree (given by the accrediting body, The Institute of Physics).

Concerning teaching/learning activities, I explored several methods to

increase the interactivity of my teaching in order to increase the active involvement of students. First, I explored simple and quick ways to involve students during lectures as well as providing feedback to me and the students about their learning. The simplest but nevertheless very effective interactive method is asking questions to students, called “interrogating practice” by [Fry et al. \(2009\)](#). The goal of questioning is to help students learn using deep approaches (searching for meaning, comparing and contrasting to existing knowledge), or using [Biggs & Tang \(2007\)](#)’s analogy, “make Roberts learn like Susans”. The other main interactive method that I explored is what [Cavanagh \(2011\)](#) calls “Lectorials” (see Sect. 5.4). The idea of “Lectorials” is to merge traditional lectures with problem classes. This is what I have done for a few years for my 2-hour mechanics sessions. I found out, while writing this portfolio, that [Cavanagh \(2011\)](#) and others had explored this merger thus providing much more statistical weight to the discussion and findings. The results of my and the other studies reveal that although it is not clear that this merger improves learning, it certainly helps maintain interest and concentration during lectures. The main point that students complained about in my lectorials is that the problem solving session were too short, “not leaving enough time for the students to get stuck in”. The other issue is that different students make progress at very different pace and there is the risk that some good students would get bored if the problem solving task was too long. One way to improve this aspect is to first deal with the “solving strategy“, which, although it does not take too much time, is the part most students struggle with. Then this strategy could even be discussed between small groups of students and only the fastest students would actually start solving the problems in details. Many students would like to still have traditional problem classes on top of the lectorials, demonstrating the popularity of problems classes. This popularity is explained by the fact that students are in charge of their learning and also the opportunity for students to discuss the problems with peers and staff members.

Concerning exams, we could complain that students are mostly concerned about whether or not a topic is the exams in order to decide if they are going to invest time in learning the topic. But complaining does not

help. Biggs & Tang (2007) explain that we have to (constructively) align assessment tasks with the ILOs. When this alignment is in place, students aims and module goals (ILOs) are the same and we can get students to learn what and how we want them to. In Chapter 6, I first reviewed the various assessment tasks at both the degree and module levels and checked that they are aligned with the ILOs. Then I focused on two main assessment tasks, a formative, assessed problems sheets, and a summative, the exam. The assessed problems sheets and the accompanying comments given to the students provide feedback to students on how well they have achieved the ILOs on the part of the module covered so far. This thus provides them with the opportunity to improve their learning before the final assessment, the exam. Concerning the exam, I have checked how different exam questions assess different ILOs in order to assess all the ILOs. This ensures that students can only obtain a good grade if they have achieve all ILOs.

Throughout the portfolio, I have discussed feedback and found that even though feedback is a very good idea in principle, it is not easy to make it work in practice. For example, it was not easy to obtain feedback on my teaching methods from the students. Either my feedback forms were too long and I only received a few forms back or the forms were too general and I did not obtain feedback on a specific aspect of my teaching I wanted feedback on (e. g. use online resources). Feedback to students is also not always as effective as we might think. I discussed in Sect. 6.3.1 the lower-than-expected fraction of students who came to collect their exam papers following a student-led campaign on exam feedback. We also try to improve the feedback accompanying assessed problems sheets in our department by providing model answers as soon as possible after submission deadlines in the hope that more students will check what they have done right or wrong while the topic is still on their mind. This was not easy to put in place since students were allowed extensions for such work and we had to abolish these extensions for the new system to work.

An unintended learning outcome that came out of this course is to experience learning from a student's perspective. I finished my undergraduate studies more than ten years ago and it is easy to forget how it is to be a student. I also learnt more on the students perspective by teaching not only

undergraduates but also future or practising school teachers. It was humbling for me to discover how hard it was for me to keep up with the TLHEP assignments let alone get really started with the course, even though I had decided to start the course myself. It made me more aware of the struggles the students may go through and how supportive feedback can help the students keep their motivation and hope of success as discussed in Sect. 3.1. Even though we rarely talk about the emotional aspect of learning, it is clear that it plays a role in learning and I would like to explore this aspect more in the future. In particular, I would like to become as effective as possible in helping the students overcome potential “psychological barriers” that they encounter for example with mathematics. Another aspect of learning that I am interested in is the transformative effect learning has on learners (explored for example by [Mezirow, 1991](#)).

As I said above, I have learnt a lot during this course but it only represents the start of my journey of reflective practice. Year after year, I will strive to check the alignment between the different parts of my teaching, explore new methods and learning activities in order to improve my teaching and evaluate their impact using feedback and learning theories such as constructivism. I give here a non-exhaustive list of improvements that I plan to implement in the near future:

- Be more aware of how the feedback I give to students affect their learning and in particular their motivation. Students’ confidence and motivation are especially affected by feedback at the start of a module when they are overwhelmed with a new topic and possibly a whole new environment like my SASP students. I will for example highlight that their probability of success depends more on the work they will do than on their prior knowledge. I will also do my best to provide supportive and encouraging feedback when distributing assessed work back to the students and re-iterate at that time that I am here to help them.
- In chapter 4, I improved my ILOs for the EM module using the recommendations given in [Biggs & Tang \(2007\)](#). I therefore plan to use these new ILOs for next year and also to revise my ILOs for my other

modules in a similar way. I also plan to use these ILOs to explain to the students what I expect of them and the fact that they can use the ILOs to evaluate how well they are learning and to establish a revision strategy.

- As discussed above, feedback is not easy to implement and I have learnt how to improve feedback forms. I will thus shorten the feedback forms that I use for the SASP and SKE modules so that I can get feedback from more students. On the other hand, I will prepare a few specific questions for the EM students in order to obtain feedback on the new methods I will introduce next year (new handouts, screencasts).
- I will definitely improve my EM module slides. Since this was the first year that I taught this module, it is not a surprise that I did not get it right the first time and using my own reflection and students feedback, I am now in a good position to improve them. In particular, I will try to reduce the content written on the slides during lectures by preparing a second version of lecture slides that will help me better balance the use of PowerPoint and white board. I will also attempt to use interactive white boards in the near future. I will however not prepare lecture notes since I think it is a valuable skill for the students to use textbooks to support their learning. I hope that by adding it to my ILOs and explaining it clearly to the students, they will take this opportunity of developing their self-directed learning skills.

Bibliography

- Anderson, L. W., & Krathwohl, D. R. (2001) *A Taxonomy for Learning, Teaching, and Assessing: A Revision of Bloom's Taxonomy of Educational Objectives*, New York: Longman
- Atherton, J. S. (2011) *Learning and Teaching; Assimilation and Accommodation* [On-line: UK] retrieved on 30 June 2011 from <http://www.learningandteaching.info/learning/assimacc.htm>
- Biggs, J., & Tang, C. (2007) *Teaching for Quality Learning at University*, 3rd Edition, Maidenhead, UK: The Society for Research into Higher Education & Open University Press, McGraw-Hill
- Bloom, B. S. (1979) *Taxonomy of Educational Objectives*, Handbook 1, *The cognitive domain*, New York: David McKay
- Bligh, D. A. (1972) *What's the use of lecture?*, Harmondsworth: Penguin
- Bruner, J. S. (1960) *The Process of Education*, Cambridge, MA: Harvard University Press.
- Cavanagh, M. (2011) "Students' experiences of active engagement through cooperative learning activities in lectures" in *Active Learning in Higher Education*, vol. 12 no. 1 23-33, March
- Cousin, G. (2006) *An Introduction to threshold concepts*, HEA Planet No. 17, December
- Fry, H., Ketteridge, S., & Marshall, S. (2009) *A HANDBOOK FOR Teaching and Learning in Higher Education*, 3rd Edition, New York: Routledge

Huxham, M. (2005) "*Learning in lectures*" in *Active Learning in Higher Education*, vol. 6, 17-31

The Institute of Physics, (2009) *The Physics Degree: Graduate Skills Base and the Core of Physics* [On-line: UK] retrieved from http://www.iop.org/education/higher_education/accreditation/file_43311.pdf

Mezirow, J. (1991) *Transformative Dimensions of Adult Learning*, San Francisco, CA: Jossey-Bass.

The Quality Assurance Agency for Higher Education, (2008) *physics, astronomy and astrophysics*, ISBN 978 1 84482 805 0, retrieved from <http://www.qaa.ac.uk/Publications/InformationAndGuidance/Pages/Subject-benchmark-statement-Physics-astronomy-and-astrophysics.aspx>

Ramsden, J. (2003) *Learning to Teach in Higher Education*, London, UK: RoutledgeFalmer.

Vygotsky, L. (1978) *Mind in Society: The development of higher Psychological Processes*, Cambridge, MA: Harvard University Press.